ControlTrace: Teaching Claude the Difference Between Allegation and Proof

Anthropic Conceptual Reasoning Fellowship Proposal (Jan – May 2027) • Ram Bharathan

Public Pilot Repository & Scorer: https://github.com/ramtoo-cell/controltrace • **Standards Anchor:** PCAOB AS 1105 & ISA 500

Question. When Claude investigates a compliance matter, does it reach conclusions on sufficient, appropriate evidence, and does it escalate because the evidence demands it or because the allegation sounds serious? Outcome-only benchmarks cannot answer this; a model can be right while leaning on material it should have excluded.

Approach. Ground truth is anchored in the profession's codified standards for audit evidence (PCAOB AS 1105, ISA 500): evidence must be relevant and reliable, independent sources outrank management assertions, and an allegation is not corroboration. Each case gives an agent a task and an evidence corpus containing decisive items, red herrings, and misleading items, and scores three things separately: conclusion, evidence faithfulness, and escalation.

Starting point. Six pilot cases with named traps (*salience*, *accusation-is-not-evidence*, *calibrated uncertainty*). A blind Claude baseline reached the right outcome in 5 of 6 cases, cited evidence that should have been excluded in 5 of 6, and in one case (CT-001) confirmed a violation on an unsubstantiated complaint after computing that a vendor's disclosure had lapsed >3 years earlier. Limits: six cases, one model, one sample per case, ground truth not yet independently validated.

Plan (about 17 weeks)

- **Weeks 1–3, validity:** Formalize the rubric against PCAOB AS 1105 and ISA 500; have 3 independent audit professionals label ground truth blind, with a discuss-then-revise round; keep only high-agreement cases.
- **Weeks 4–9, scale:** Grow to 15–20 cases across audit, regulatory, and legal judgment; build minimal pairs that differ by one piece of evidence so the correct action flips between resolve and escalate.
- **Weeks 10–13, measurement:** Multiple samples across Claude and other model families; results by trap type; any model grader calibrated against expert labels.
- **Weeks 14–17, delivery:** Package as an internal evaluation and, with mentors, as RL environments rewarding faithfulness and calibrated escalation; write up results, negative findings included.

Deliverables. A validated case suite with documented expert agreement; a scorer with known failure modes; a multi-model report; an RL-ready version; a public write-up if Anthropic approves.

Risks. Contested ground truth: drop the case rather than force it. Contamination and gaming: synthetic, held-out cases varied in surface form (with held-out ground truth kept private via SHA-256 commitments). Confidentiality: every case is 100% synthetic, with zero employer or client data.

Cover Note — Anthropic Conceptual Reasoning Fellowship

Ram Bharathan • October 2026

Dear Anthropic Fellows team,

Auditors learn early that the most dangerous error is not the wrong conclusion but the right one reached for the wrong reasons. It passes review, and it fails precisely when the facts shift. I believe the same holds for AI agents, and that it is the failure accuracy-only evaluation is least equipped to see.

In my pilot benchmark (https://github.com/ramtoo-cell/controltrace), a Claude model correctly computed that a vendor's disclosure had lapsed more than three years earlier, then treated an unsubstantiated complaint from a competitor as proof of a violation and escalated it. An experienced auditor would have closed the matter. Across six cases the model usually reached the right outcome, but often while citing evidence it should have excluded.

The audit profession has spent decades codifying when evidence is sufficient and appropriate. I want to turn that codified judgment into evaluations and RL environments that teach Claude the difference between an allegation and proof, and between a matter to resolve and one to escalate.

Why Anthropic: Anthropic's work on Constitutional AI, process faithfulness, and scalable oversight treats reasoning integrity as a core safety problem rather than a post-hoc filter. Because Claude is increasingly relied upon in legal, regulatory, and enterprise compliance workflows where questions have no simple answer key, working alongside Anthropic researchers to test and train how Claude weighs contested evidence is where nearly two decades of audit judgment can do the most good.

Ram Bharathan